



MsAlisia Phase 1 MVP Build Package

Frontend + Backend + Documentation - Grades 3-6, assessment engine, adaptive tutoring, Claude primary / Groq fallback

Area	Included in Package
Frontend	React + TypeScript + Vite interface designed for non-technical parents, students, and admins.
Backend	Python FastAPI backend with Claude-first and Groq-fallback LLM router.
Curriculum	Grades 3-6 support across Math, ELA, and Writing with subject-level progression.
Assessment	Math, ELA, and Writing assessment evaluation that returns competency level, gaps, strengths, and progression.
Future Modules	Voice, mobile, teacher portal, school integrations, advanced analytics, full K-12, and advanced handwriting shown as Coming Soon.

1. Executive Summary

This package provides the Phase 1 MVP scaffold for MsAlisia. The core goal is to create a stable, simple, user-friendly learning platform that supports Grades 3-6, mandatory assessments, adaptive progression by subject, and guided tutoring with Ms Alisia. The interface is intentionally non-technical and avoids heavy AI branding while still using an LLM backend for tutoring and evaluation.

The backend supports Claude as the preferred model and Groq as fallback. This lets development continue while the client is still preparing the Claude API key. If Claude is present, it is used first. If Claude is missing or fails and fallback is enabled, Groq is used.

2. Phase 1 Functional Scope

- Grades 3-6 curriculum support, replacing the earlier Grades 3-5 scope.
- Adaptive subject-level progression, so each student can work above or below enrolled grade by subject.
- Mandatory Math, ELA, and Writing assessments for MVP launch.
- Assessment output includes competency level, strengths, learning gaps, and recommended progression.
- Writing support covers grammar, sentence structure, organization, comprehension, and composition.
- Handwriting/penmanship MVP support through worksheet or image upload with lightweight AI feedback.
- Simple parent onboarding and profile context for personalization.
- Parent reports, admin visibility, and future modules shown as Coming Soon.

3. Frontend Overview

The frontend is built with React, TypeScript, and Vite. It uses a calm lilac/purple and gold design language and presents all major workflows as large, clear cards and simple navigation items.

Screen	Purpose
Home	Starting dashboard with student snapshot and recommended next step.
Parent Setup	Collects student name, grade, confidence style, focus notes, and parent notes.
Assessments	Runs Math, ELA, and Writing quick checks for Grades 3-6.
Start Learning	Ms Alisia chat interface with subject and topic controls.
Homework & Handwriting	Upload workflow and lightweight feedback area for homework, writing, and penmanship.
Reports	Parent-friendly subject levels and next steps.
Admin	Non-technical admin visibility for users, assessments, AI usage, and safety review.
Future Modules	Coming Soon cards for later phases without full functionality.

4. Backend Overview

The backend is a FastAPI service. It includes API routes, data models, SQLite storage for development, curriculum definitions, prompts, assessment evaluation, and the LLM provider router.

File	Purpose
app/main.py	FastAPI application, CORS, health check, curriculum, student, chat, assessment, homework, admin, and future module routes.
app/config.py	Environment configuration for CORS, database path, Claude, Groq, fallback behavior, tokens, and temperature.
app/models.py	Pydantic models for students, chat, assessments, and homework feedback.
app/curriculum.py	Grades 3-6 Math, ELA, and Writing topic maps plus handwriting rubric.
app/prompts.py	Ms Alisia safety, tutoring, subject, assessment, and homework prompt instructions.
app/services/llm/router.py	Claude-first and Groq-fallback routing logic.
app/services/assessment_service.py	Assessment evaluation, JSON parsing, and database save.
app/database.py	SQLite schema and helper functions for prototype persistence.

5. LLM Routing Design

The backend focuses heavily on LLM correctness. The prompt layer is course-aware and includes curriculum context, subject-specific rules, safety constraints, and the required child-friendly tutoring format.

Environment variables:

```
PRIMARY_LLM_PROVIDER=c laude
FALLBACK_LLM_PROVIDER=groq
ANTHROPIC_API_KEY=<client key when provided>
GROQ_API_KEY=<developer fallback key>
FALLBACK_ON_LLM_ERROR=true
```

Routing behavior:

- If ANTHROPIC_API_KEY is present, Claude is used as the primary model.
- If Claude API key is missing, the system automatically uses Groq.
- If Claude fails and fallback is enabled, the system uses Groq.
- If no API keys are available, a local fallback keeps the interface usable for development only.
- LLM events are recorded to support future admin usage visibility.

6. Course-Aware Prompt Strategy

Subject	LLM Focus
Math	Reasoning, number sense, short worked steps, one quick practice question, no answer dumping.
ELA	Reading comprehension, vocabulary, grammar, text evidence, inference, and short reading checks.
Writing	Grammar, sentence structure, organization, clarity, comprehension, and composition.
Handwriting	Lightweight feedback on legibility, spacing, neatness, letter formation, and overall readability through upload workflow.

The model is instructed to keep responses short, start with encouragement, explain one small idea, ask exactly one validation question, and avoid long instructional articles.

7. Assessment Engine

The assessment endpoint accepts questions and student answers, sends them through the course-aware assessment prompt, and expects concise JSON with the following output.

- estimated_level
- score_label
- strengths
- learning_gaps
- recommended_progression
- parent_summary

The result can update each subject independently. For example, one student can be Grade 5 in Math, Grade 4 in ELA, and Grade 3 in Writing.

8. Coming Soon / Future Modules

Phase 1 shows future modules in the UI without building full functionality. This communicates the complete product vision while keeping the MVP focused on launch-critical work.

- Voice Learning
- Mobile App
- Teacher Portal
- School/LMS Integrations
- Advanced Analytics
- Full K-12 Expansion
- Advanced Handwriting AI
- Science
- Social Studies
- Test Prep

9. Running the Project Locally

Backend:

```
cd backend
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
cp .env.example .env
uvicorn app.main:app --reload --port 8000
```

Frontend:

```
cd frontend
npm install
cp .env.example .env
npm run dev
```

Open <http://localhost:5173> for the app and <http://localhost:8000/docs> for the API documentation.

10. Production Notes

- Replace SQLite with PostgreSQL or Supabase/Postgres for production.
- Add real authentication and authorization before live users.
- Add rate limiting and abuse prevention for public endpoints.
- Move secrets to deployment environment variables or a secret manager.
- Add real file storage for uploaded homework and writing samples.
- Add Stripe only when payment validation is ready.
- Add safety review workflows and parent escalation rules before launch.

11. Package Contents

The ZIP package includes /frontend, /backend, /docs, root README, docker-compose.yml, and this PDF documentation. Each side also has its own README and .env.example file.